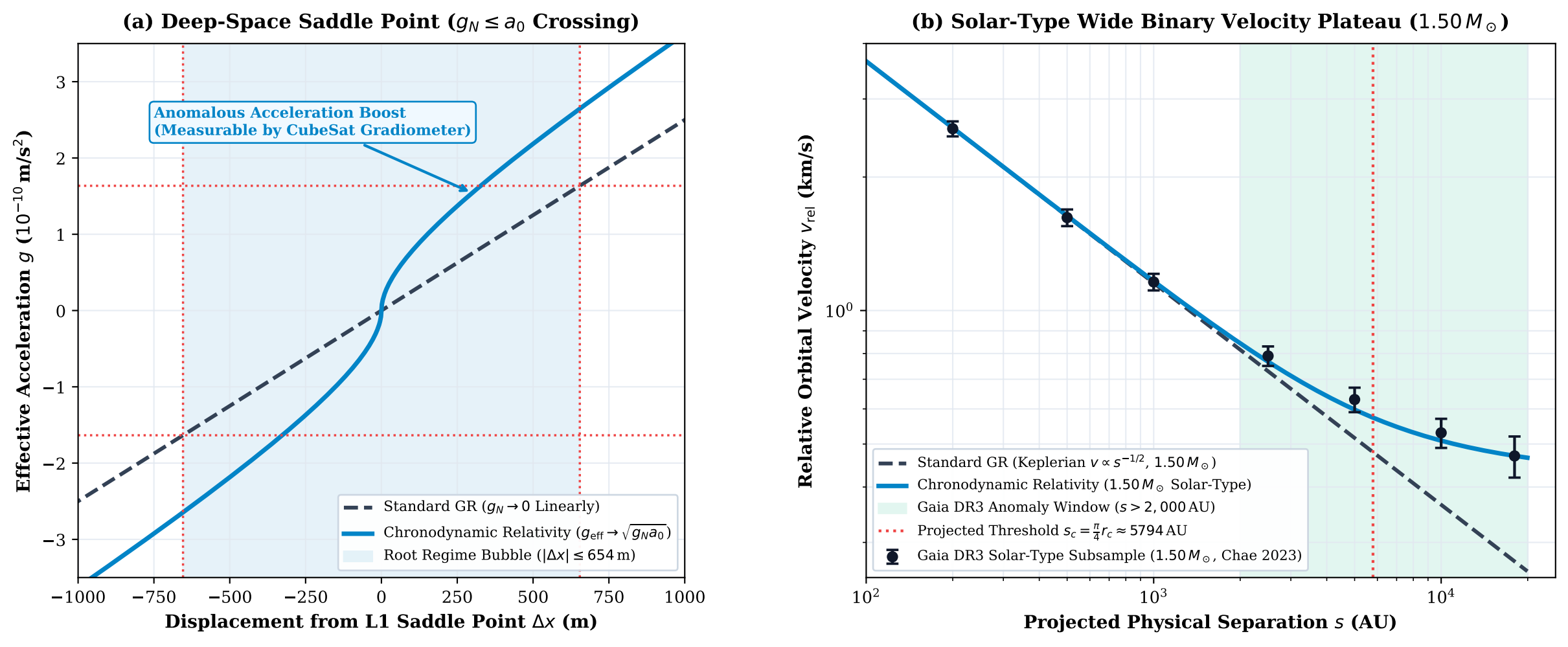
(a) Deep-Space Saddle Point ($g_N \leq a_0$ Crossing)
Anomalous Acceleration Boost
(Measurable by CubeSat Gradiometer)
Standard GR ($g_N \to 0$ Linearly)
Chronodynamic Relativity ($g_{\text{eff}} \to \sqrt{g_N a_0}$)
Root Regime Bubble ($|\Delta x| \leq 654$ m)
Effective Acceleration g (10^{-10} m/s^2)
Displacement from L1 Saddle Point Δx (m)
3
2
1
0
−1
−2
−3
−1000
−750
−500
−250
0
250
500
750
1000
(b) Solar-Type Wide Binary Velocity Plateau ($1.50\,M_\odot$)
Standard GR (Keplerian $v \propto s^{-1/2}$, $1.50\,M_\odot$)
Chronodynamic Relativity ($1.50\,M_\odot$ Solar-Type)
Gaia DR3 Anomaly Window ($s > 2,000$ AU)
Projected Threshold $s_c = \frac{\pi}{4} r_c \approx 5794$ AU
Gaia DR3 Solar-Type Subsample ($1.50\,M_\odot$, Chae 2023)
Relative Orbital Velocity v_{rel} (km/s)
Projected Physical Separation s (AU)
10^0
10^2
10^3
10^4